

1. a. $V(x) = (Px) \cdot x$

$$\begin{aligned} \frac{d}{dt} V(x) &= \frac{d}{dt} (Px) \cdot x + Px \cdot \frac{d}{dt} x \\ &= P(Ax) \cdot x + Px \cdot Ax \\ &= P(Ax) \cdot x + A^T P x \cdot x \\ &= (PAx + A^T P x) \cdot x \\ &= (PA + A^T P)x \cdot x \end{aligned}$$

Lyapunov iff | Proof

$$\begin{aligned} V(x^*) &= 0 & x^* &= 0 \text{ due to linearity: } V(0) = P \cdot 0 \cdot 0 = 0 \\ \text{and } V(x^* + \epsilon) &> 0 & V(\epsilon) &= P \cdot \epsilon \cdot \epsilon > 0 \text{ (iff } P \text{ is positive definite)} \\ \text{and } \frac{d}{dt} V(x(t)) &< 0 & \frac{d}{dt} V &= (PA + A^T P)x \cdot x < 0 \text{ (iff } PA + A^T P \text{ is negative definite)} \end{aligned}$$

ie A is positive definite

b) For all eigenvalues and vectors of a positive definite symmetric matrix M λ_i and V_i ,

$$M V_i = \lambda_i V_i$$

$$(M V_i) \cdot V_i = (\lambda_i V_i) \cdot V_i$$

$$V_i \cdot (M V_i) = \lambda_i (V_i \cdot V_i)$$

$$V_i^T M V_i = \lambda_i (V_i \cdot V_i)$$

> 0

$$\Rightarrow 0 < \frac{V_i^T M V_i}{V_i \cdot V_i} = \lambda_i$$

so (all eigenvalues are positive)

$$\lambda_{\min}(M) \|x\|^2 \leq Mx \cdot x \leq \lambda_{\max}(M) \|x\|^2$$

$$V(x) = Px \cdot x \leq \lambda_{\max}(P) \|x\|^2$$

$$\Rightarrow \|x\|^2 \leq \frac{V(x)}{\lambda_{\max}(P)}$$

$$-\frac{d}{dt} V(x(t)) = Qx \cdot x \geq \lambda_{\min}(Q) \|x(t)\|^2 \geq \frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V(x)$$

$$\frac{dV}{dt} \leq - \underbrace{\frac{\lambda_{\min} Q}{\lambda_{\max} P}}_{\text{Lipschitz constant}} V$$

$$\|x\|^2 \lambda_{\min} P \leq \|v\| \leq \exp\left(-\frac{\lambda_{\min} Q}{\lambda_{\max} P} t\right) \|v_0\|$$

$$\|v_0\| \leq \|x_0\|^2 \lambda_{\max} P$$

$$\|x\|^2 \leq \frac{\lambda_{\max} P}{\lambda_{\min} P} \exp\left(-\frac{\lambda_{\min} Q}{\lambda_{\max} P} t\right) \|x_0\|^2$$

$$\|x\| \leq \sqrt{\frac{\lambda_{\max} P}{\lambda_{\min} P}} \exp\left(-\frac{\lambda_{\min} Q}{2 \lambda_{\max} P} t\right) \|x_0\|$$

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$$\text{Therefore } A x \cdot x = \sum A a_i v_i \cdot a_i v_i = \sum \lambda a_i v_i \cdot v_i = \sum \lambda \underbrace{(a_i v_i)^2}_{\substack{\text{negative} \quad \text{positive}}} \leq 0$$

eigenvalues, which exist due to diagonalizability

Therefore, A is negative definite

$$\text{let } M(t) = \exp(tA^T) Q \exp(tA)$$

$$\frac{dM}{dt} = A^T \exp(tA^T) Q \exp(tA) + \exp(tA^T) Q \exp(tA) A = A^T M + M A$$

$$M(t) = \int_0^t \frac{dM}{dt} dt + M(0) = A^T \underbrace{\int_0^t M dt}_P + \underbrace{\left(\int_0^t M dt \right) A}_P$$

$$M(\infty) = A^T P + P A + M(0) = \exp(\infty A^T) Q \exp(\infty A) + M(0)$$

$$= \underbrace{\exp(\infty A)^T}_P Q \exp(\infty A) + M(0)$$

Since A is negative definite, $\lim_{t \rightarrow \infty} \exp(tA) = 0$

$$= M(0) = \exp(0) Q \exp(0) = I Q I = Q$$

Therefore, $(A^T P + P A = Q)$

2) ie $G(0) = G(1) = 0$
 has cusp at ξ w/ $\Delta G'(\xi) = -1$
 $-G'' + pG' + qG = \delta_\xi$

Let any sufficiently
 squiggly line denote
 the Greek letter xi

a) $\frac{dw}{dx} = p(x)w(x)$

$$\int \frac{dw}{w} = \int p(x) dx$$

$$w = w_0 e^{\int p(x) dx}$$

$$\Rightarrow \exists x_0, w(x_0) = 0 \Leftrightarrow w_0 = 0 \Leftrightarrow \forall x, w(x) = 0$$

Assume for the sake of contradiction that $w(x_0) = 0$

$\Rightarrow w(x) = 0$ for all $x \Rightarrow u_1$ and u_2 are linearly dependent

$$\Rightarrow \exists k \forall x, u_1(x) = k u_2(x)$$

$$u_1(0) = k u_2(0) = 0 \Rightarrow k = 0 \text{ or } u_2(0) = 0$$

If $k = 0$, then $u_1(0) = 0$ is a contradiction. Therefore let $k \neq 0$

If $u_2(0) = 0$, then by uniqueness $u_2(x) = 0$ generally.

Therefore $u_2(0) = 0$, which is a contradiction.

Therefore $w(0) \neq 0 \Rightarrow \forall t, w(t) \neq 0$

b) $G(0, \xi) = \frac{-\overset{=0}{u_1(0)} u_2(\xi)}{w(\xi)} \neq 0$

$$G(1, \xi) = \frac{-\overset{=0}{u_1(\xi)} u_2(1)}{w(\xi)} \neq 0$$

$$\text{if } x < \xi: -\frac{d^2}{dx^2} \left(\frac{u_1(x) u_2(\xi)}{w(\xi)} \right) + p(x) \frac{d}{dx} \left(\frac{u_1(x) u_2(\xi)}{w(\xi)} \right) + \frac{q(x) u_1(x) u_2(\xi)}{w(\xi)} = \frac{u_2(\xi)}{w(\xi)} \left(u_1''(x) + p(x) u_1'(x) + q(x) u_1(x) \right) = \frac{u_2(\xi)}{w(\xi)}(0) \neq 0$$

$$\text{if } x > \xi: -\frac{d^2}{dx^2} \left(\frac{u_2(x) u_1(\xi)}{w(\xi)} \right) + p(x) \frac{d}{dx} \left(\frac{u_2(x) u_1(\xi)}{w(\xi)} \right) + \frac{q(x) u_2(x) u_1(\xi)}{w(\xi)} = \frac{u_1(\xi)}{w(\xi)} \left(u_2''(x) + p(x) u_2'(x) + q(x) u_2(x) \right) = \frac{u_1(\xi)}{w(\xi)}(0) \neq 0$$

$$\lim_{x \rightarrow \xi^+} G(x, \xi) = \frac{u_1(\xi) u_2(\xi)}{w(\xi)}$$

$$\lim_{x \rightarrow \xi^-} G(x, \xi) = \frac{u_1(\xi) u_2(\xi)}{w(\xi)}$$

equal, therefore Continuous!

$$\lim_{x \rightarrow \xi^+} G'(x, \xi) = \frac{u_1(\xi) u_2'(\xi)}{w(\xi)}$$

$$\lim_{x \rightarrow \xi^-} G'(x, \xi) = \frac{u_1'(\xi) u_2(\xi)}{w(\xi)}$$

$$\frac{u_1(\xi) u_2'(\xi)}{w(\xi)} - \frac{u_1'(\xi) u_2(\xi)}{w(\xi)} = \frac{u_1(\xi) u_2'(\xi) - u_1'(\xi) u_2(\xi)}{w(\xi)} = \frac{-w'(\xi)}{w(\xi)} \neq -1$$

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$$\begin{aligned} -u'' + pu' + qu &= f \\ -v'' + pv' + qv &= f \\ \hline -(u-v)'' + p(u-v)' + q(u-v) &= 0 \end{aligned}$$

therefore, $u-v$ solves the homogeneous system.

by uniqueness, the only solution to the homogeneous system is 0. therefore, $u-v=0$

$$\Rightarrow v=u \quad \Rightarrow u \text{ is unique}$$

$$\textcircled{d} u(x) = \int_0^x G(x, \xi) f(\xi) d\xi = \int_0^x -\frac{u_1(\xi)u_2(x)f(\xi)}{w(\xi)} d\xi + \int_x^1 -\frac{u_1(x)u_2(\xi)f(\xi)}{w(\xi)} d\xi$$

$$-u'' + pu' + qu = f$$

$$= u_2(x) \int_0^x -\frac{u_1(\xi)f(\xi)}{w(\xi)} d\xi + u_1(x) \int_x^1 -\frac{u_2(\xi)f(\xi)}{w(\xi)} d\xi$$

$$u'(x) = u_2'(x) \int_0^x -\frac{u_1(\xi)f(\xi)}{w(\xi)} d\xi + u_2(x) \left(-\frac{u_1(x)f(x)}{w(x)} \right) + u_1'(x) \int_x^1 -\frac{u_2(\xi)f(\xi)}{w(\xi)} d\xi + u_1(x) \left(-\frac{u_2(x)f(x)}{w(x)} \right)$$

$$= u_2'(x) \int_0^x -\frac{u_1(\xi)f(\xi)}{w(\xi)} d\xi + u_1'(x) \int_x^1 -\frac{u_2(\xi)f(\xi)}{w(\xi)} d\xi$$

$$u'' = u_2''(x) \int_0^x -\frac{u_1(\xi)f(\xi)}{w(\xi)} d\xi + u_2' \left(-\frac{u_1(x)f(x)}{w(x)} \right) + u_1'' \int_x^1 -\frac{u_2(\xi)f(\xi)}{w(\xi)} d\xi + u_1' \left(-\frac{u_2(x)f(x)}{w(x)} \right)$$

$$= u_2''(x) \int_0^x -\frac{u_1(\xi)f(\xi)}{w(\xi)} d\xi + u_1'' \int_x^1 -\frac{u_2(\xi)f(\xi)}{w(\xi)} d\xi + \frac{f(x)}{w(x)} (u_1' u_2 - u_2' u_1)$$

$$-u'' + pu' + qu = -u_2''(x) \int_0^x -\frac{u_1(\xi)f(\xi)}{w(\xi)} d\xi - u_1'' \int_x^1 -\frac{u_2(\xi)f(\xi)}{w(\xi)} d\xi + f(x) + p u_2'(x) \int_0^x -\frac{u_1(\xi)f(\xi)}{w(\xi)} d\xi$$

$$+ p u_1'(x) \int_x^1 -\frac{u_2(\xi)f(\xi)}{w(\xi)} d\xi + q u_2(x) \int_0^x -\frac{u_1(\xi)f(\xi)}{w(\xi)} d\xi + q u_1(x) \int_x^1 -\frac{u_2(\xi)f(\xi)}{w(\xi)} d\xi$$

$$-u'' + pu' + qu = \left(\int_0^x -\frac{u_1(\xi)f(\xi)}{w(\xi)} d\xi \right) \underbrace{(-u_2'' + p u_2' + q u_2)}_{=0} + \left(\int_x^1 -\frac{u_2(\xi)f(\xi)}{w(\xi)} d\xi \right) \underbrace{(-u_1'' + p u_1' + q u_1)}_{=0} + f(x)$$

$$-u'' + pu' + qu = f(x)$$

$\boxed{2e}$ $-u_1'' + 2u_1 = 0 \Rightarrow u_1' = 2u_1$
 $u_2'' = 2u_2$

$$W' = \frac{d}{dx}(u_1' u_2 - u_1 u_2') = u_2 u_1'' - u_1 u_2'' = u_2 2u_1 - u_1 2u_2 = 0$$

$\Rightarrow W$ is constant, denoted W_0

either $x \geq \xi$ or $x < \xi$

If $x \geq \xi$, then $G(x, \xi) = -\frac{u_1(\xi)u_2(x)}{W(\xi)} = -\frac{u_1(\xi)u_2(x)}{W_0} = -\frac{u_1(\xi)u_2(x)}{W(x)} = G(\xi, x)$

If $\xi \geq x$, then $G(x, \xi) = -\frac{u_1(x)u_2(\xi)}{W(\xi)} = -\frac{u_1(x)u_2(\xi)}{W_0} = -\frac{u_1(x)u_2(\xi)}{W(x)} = G(\xi, x)$

Therefore, $G(x, \xi) = G(\xi, x)$

$\boxed{f}$ $u_1'' = 0, u_1(0) = 0, u_1'(0) = 1$

is an IVP with unique solution

$$u_1(x) = x$$

$$u_2'' = 0, u_2(1) = 0, u_2'(1) = 1$$

is an IVP with unique solution

$$u_2(x) = x - 1$$

$$W(0) = \overset{-1}{u_1' u_2} - u_1' u_2' = 0 - (-1) = 1$$

$$G(x, \xi) = \begin{cases} -x(\xi - 1) & \text{if } x \leq \xi \\ -\xi(x - 1) & \text{if } x \geq \xi \end{cases}$$

consider $f(x) = x$

$$u(x) = \int_0^1 G(x, \xi) f(\xi) d\xi = \int_0^x -\xi u_1(\xi) u_2(x) d\xi + \int_x^1 -\xi u_1(x) u_2(\xi) d\xi$$

$$= \int_0^x -\xi \xi(x-1) d\xi + \int_x^1 -\xi x(\xi-1) d\xi = \left(\frac{x^3}{3} - \frac{x^4}{4}\right) + \left(\frac{x}{6} - \frac{x^3}{2} + \frac{x^4}{3}\right) = \frac{1}{6}(x - x^3)$$

$$u(x) = \frac{1}{6}(x - x^3)$$

$$-\frac{d^2}{dx^2} u = \frac{1}{6} \left(\frac{1}{6} - \frac{1}{2} x^2 \right) = -x = -f(x) \quad \checkmark$$

$$u(0) = 0 \quad \checkmark$$

$$u(1) = \frac{1}{6}(1 - 1) = 0 \quad \checkmark$$

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a

$$\frac{1}{2\pi} \int_0^{2\pi} e^{mx} e^{-nx} dx = \frac{1}{2\pi} \int_0^{2\pi} e^{(m-n)x} dx$$

if $m=n \rightarrow \frac{1}{2\pi} \int_0^{2\pi} dx = 1$

if $m \neq n \rightarrow \frac{1}{2\pi} \int_0^{2\pi} e^{(m-n)x} dx = 0 - 0 = 0$

$$= \begin{cases} 1 & \text{if } m=n \\ 0 & \text{if } m \neq n \end{cases}$$

$$C_m = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-mx} dx = \frac{1}{2\pi} \int_0^{2\pi} f(x) (\cos(mx) - i \sin(mx)) dx$$

if $m > 0 \rightarrow \frac{1}{2\pi} \int_0^{2\pi} f(x) \cos(mx) dx - i \frac{1}{2\pi} \int_0^{2\pi} f(x) \sin(mx) dx = \frac{1}{2} (a_m - i b_m)$

if $m < 0 \rightarrow \frac{1}{2\pi} \int_0^{2\pi} f(x) \cos(-mx) dx + i \frac{1}{2\pi} \int_0^{2\pi} f(x) \sin(-mx) dx = \frac{1}{2} (a_m + i b_m)$

if $m=0, C_m = \frac{1}{2\pi} \int_0^{2\pi} f(x) dx = a_0$

b

$$2\pi C_m = \int_0^{2\pi} f(x) e^{-mx} dx = \left[\frac{1}{m} f(x) e^{-mx} - \int_0^{2\pi} f'(x) e^{-mx} dx \right] = \frac{1}{m} \left((f(2\pi) \underbrace{e^{-2\pi i m}}_{=1} - f(0) e^0 - \int_0^{2\pi} f'(x) e^{-mx} dx) \right) = \frac{1}{m} \left(f(0) - \int_0^{2\pi} f'(x) e^{-mx} dx \right)$$

let F be the Fourier approximation of f

$$\forall x \quad f(x) = F(x) \Rightarrow f'(x) = F'(x)$$

$$F'(x) = \sum_{n=1}^{\infty} \frac{d}{dx} (a_n \cos(nx) + b_n \sin(nx)) = \sum_{n=1}^{\infty} (n b_n \cos(nx) + (-n a_n) \sin(nx))$$

$$a'_n = n b_n \quad b'_n = -n a_n$$

Because f is continuously differentiable, $\exists \sup_{[0, 2\pi]} |f'|$.

$$|a'_n| = \left| \frac{1}{\pi} \int_0^{2\pi} f'(x) \cos(nx) dx \right| \leq \frac{1}{\pi} \int_0^{2\pi} |f'(x) \cos(nx)| dx \leq \frac{1}{\pi} \int_0^{2\pi} |f'(x)| dx \leq \frac{1}{\pi} \cdot 2\pi \cdot \sup |f'| = 2 \cdot \sup |f'|$$

$$|b'_n| = \left| \frac{1}{\pi} \int_0^{2\pi} f'(x) \sin(nx) dx \right| \leq \frac{1}{\pi} \int_0^{2\pi} |f'(x) \sin(nx)| dx \leq \frac{1}{\pi} \int_0^{2\pi} |f'(x)| dx \leq \frac{1}{\pi} \cdot 2\pi \cdot \sup |f'| = 2 \cdot \sup |f'|$$

Therefore $\|C'_m\|$ is bounded. Denote the bound with $C'_{\max}$.

$$n |b_n| = \|a'_n\| \leq \|C'_n\| \leq C'_{\max}$$

$$n |a_n| = \|b'_n\| \leq \|C'_n\| \leq C'_{\max}$$

$$\Rightarrow \|C_n\| \leq |b_n| + |a_n| \leq \frac{1}{n} (2 C'_{\max})$$

$$\Rightarrow \|C_n\| \in O\left(\frac{1}{n}\right)$$

constant

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$$2\pi c_m = \int_0^{2\pi} f(x) e^{-mx} dx = \int_0^{2\pi} \frac{1}{m} f(x) e^{-mx} dx - \int_0^{2\pi} \frac{1}{m} f(x) e^{-mx} dx = \frac{1}{m} \left(\underbrace{f(2\pi) e^{-2\pi i m}}_{=0} - f(0) e^0 - \int_0^{2\pi} f'(x) e^{-mx} dx \right) = \frac{1}{m} \int_0^{2\pi} f'(x) e^{-mx} dx$$

$$= \frac{1}{m} \left(\int_0^{2\pi} \frac{1}{m} f'(x) e^{-mx} dx - \int_0^{2\pi} \frac{1}{m} f''(x) e^{-mx} dx \right) = \left(\frac{1}{m} \right)^2 \int_0^{2\pi} f''(x) e^{-mx} dx \dots = \left(\frac{1}{m} \right)^k \int_0^{2\pi} f^{(k)}(x) e^{-mx} dx$$

let F be the Fourier approximation of f

$$\forall x \quad F(x) = F(x) \Rightarrow F'(x) = F'(x)$$

$$F^{(k)}(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} = \sum_{n=-\infty}^{\infty} (in)^k c_n e^{inx}$$

$$\Rightarrow c_n^{(k)} = (in)^k c_n$$

Because f is k -continuously differentiable, $\exists \sup_{[0, 2\pi]} |f^{(k)}|$.

$$|a_n^{(k)}| = \left| \frac{1}{\pi} \int_0^{2\pi} f^{(k)}(x) \cos(nx) dx \right| \leq \frac{1}{\pi} \int_0^{2\pi} |f^{(k)}(x)| \cos(nx) dx \leq \frac{1}{\pi} \cdot 2\pi \cdot \sup |f^{(k)}| = 2 \cdot \sup |f^{(k)}|$$

$$|b_n^{(k)}| = \left| \frac{1}{\pi} \int_0^{2\pi} f^{(k)}(x) \sin(nx) dx \right| \leq \frac{1}{\pi} \int_0^{2\pi} |f^{(k)}(x)| \sin(nx) dx \leq \frac{1}{\pi} \cdot 2\pi \cdot \sup |f^{(k)}| = 2 \cdot \sup |f^{(k)}|$$

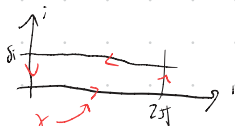
Therefore $\|c_n^{(k)}\|$ is bounded. Denote the bound with $c_{\max}^{(k)}$.

$$(in)^k \|c_n\| = \|c_n^{(k)}\| \leq c_{\max}^{(k)}$$

$$\Rightarrow \|c_n\| \leq \frac{1}{n^k} \underbrace{(2 \cdot c_{\max}^{(k)})}_{\text{constant}}$$

$$\Rightarrow \|c_n\| \in O\left(\frac{1}{n^k}\right)$$

3d helped by Jacob Greene



let $g(x) = f(x)e^{imx}$

$$0 = \int_{\gamma} g(z) dz = \int_0^S g(x) dx + \int_S^{2\pi i} g(z) dz + \int_{2\pi i}^0 g(x) dx + \int_0^0 g(x+i\delta) dx$$

$$\Rightarrow \int_0^{2\pi i} g(x) dx = \int_0^{2\pi i} g(x+i\delta) dx \Rightarrow$$

$$\left| \int_0^{2\pi i} f(x) e^{imx} dx \right| = \left| \int_0^{2\pi i} f(x+i\delta) e^{im(x+i\delta)} dx \right|$$

let $h(x) = f(x+i\delta)$

$$\int_0^{2\pi i} h(x) e^{imx} e^{-m\delta} dx = e^{-m\delta} \int_0^{2\pi i} h(x) e^{imx} dx$$

let H be Fourier approx of h , F be Fourier of f

$$H_0 := H(0), \quad H_0 = \sum c_m \Rightarrow \|c_m\| \text{ is bounded}$$

$$\|c_m\| = \left\| \int_0^{2\pi i} h(x) e^{imx} dx \right\| \Rightarrow \left\| \int_0^{2\pi i} h(x) e^{imx} dx \right\| \text{ is bounded with bound } B$$

$$\left| \int_0^{2\pi i} f(x) e^{imx} dx \right| = \left| \int_0^{2\pi i} f(x+i\delta) e^{im(x+i\delta)} dx \right| = \left\| \int_0^{2\pi i} h(x) e^{imx} e^{-m\delta} dx \right\| = \|e^{-m\delta}\| \left\| \int_0^{2\pi i} h(x) e^{imx} dx \right\| \leq \|e^{-m\delta}\| B$$

$$\Rightarrow \left| \int_0^{2\pi i} f(x+i\delta) e^{im(x+i\delta)} dx \right| \leq B e^{-m\delta}$$

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a) $f: [a, b]$

let $g(x) = f(x-a)$ $g: [0, L]$

let $L = b-a$

$$G(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2n\pi x}{L}\right) + b_n \sin\left(\frac{2n\pi x}{L}\right)$$

$a_n = \frac{2}{L} \langle g(x), \cos\left(\frac{2n\pi x}{L}\right) \rangle$
 $b_n = \frac{2}{L} \langle g(x), \sin\left(\frac{2n\pi x}{L}\right) \rangle$
 $F(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{2n\pi(x+a)}{L}\right) + b_n \sin\left(\frac{2n\pi(x+a)}{L}\right)$

$F(x) = G(x+a)$

b) $a_0 = 0$

$$a_n = \frac{2}{2\pi} \int_0^{2\pi} (x+\pi)^3 \cos\left(\frac{2n\pi x}{L}\right) dx = \int_0^{2\pi} \frac{1}{\pi} (x+\pi)^3 \cos(n\pi x) dx = \frac{1}{\pi} (x+\pi)^3 \frac{1}{n} \sin(n\pi x) - \int \frac{3}{\pi} (x+\pi)^2 \frac{1}{n} \sin(n\pi x) dx = \dots + \frac{3}{\pi} (x+\pi)^2 \frac{1}{n^2} \cos(n\pi x) - \int \frac{6}{\pi} (x+\pi) \frac{1}{n^2} \cos(n\pi x) dx$$

$$= \dots + \dots + \frac{6}{\pi^2} (x+\pi)^2 \frac{1}{n^2} \sin(n\pi x) - \int \frac{6}{\pi} \frac{1}{n^2} \sin(n\pi x) dx = \left[\frac{1}{\pi} (x+\pi)^3 \frac{1}{n} \sin(n\pi x) + \frac{3}{\pi} (x+\pi)^2 \frac{1}{n^2} \cos(n\pi x) + \frac{6}{\pi^2} (x+\pi) \frac{1}{n^2} \sin(n\pi x) - \frac{6}{\pi^2} \frac{1}{n^2} \cos(n\pi x) \right]_0^{2\pi} = 0$$

by Wirham Alphon

$$b_n = \frac{2}{2\pi} \int_0^{2\pi} (x+\pi)^3 \sin\left(\frac{2n\pi x}{L}\right) dx = \frac{12\pi^2}{n^3} - \frac{2\pi^2}{n}$$

$$f(x) = \sum_{n=1}^{\infty} \left(\frac{12\pi^2}{n^3} - \frac{2\pi^2}{n} \right) \sin(n(x-\pi))$$

c)

Consider $g(x) = x^3 - \pi^2 x$ on $[-\pi, \pi]$

Alternatively, consider $f(x) = (x-\pi)^3 - \pi^2(x-\pi)$ on $[0, 2\pi]$

f is odd $\Rightarrow a_n = 0 \forall n$

by Wirham Alphon

$$b_n = \frac{2}{2\pi} \int_0^{2\pi} ((x-\pi)^3 - \pi^2(x-\pi)) \sin\left(\frac{2n\pi x}{2\pi}\right) dx = \frac{12}{n^3}$$

let F be the Fourier approx of f

$$F(x) = \sum_{n=1}^{\infty} \frac{12}{n^3} \sin(n\pi x)$$

$F(\frac{\pi}{2}) = f(\frac{\pi}{2}) = (-\frac{\pi}{2})^3 - \pi^2(-\frac{\pi}{2}) = -\frac{\pi^3}{8} + \frac{\pi^3}{2} = \frac{3}{8}\pi^3$

$F(\frac{\pi}{2}) = \sum_{n=1}^{\infty} \frac{12}{n^3} \sin(n\pi \frac{\pi}{2}) = \sum_{k=1}^{\infty} \frac{12}{(2k)^3} \sin(k\pi) + \sum_{k=1}^{\infty} \frac{12}{(2k-1)^3} \sin(k\pi + \frac{\pi}{2}) = 12 \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{(2k-1)^3}$

even $n = 0$
odd n

$$\Rightarrow \frac{\pi^3}{32} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)^3}$$